

Week 9 – Workshop

Geothermics: Transient Heat Flow



Geophysics 3A: Potential Fields & Geothermics

July 23, 2026

Duration: 3 hours

Setting: Workshop room

Preparation

Pre-reading: Course Notes, Ch. 11

Lecture videos:

Notes:

Purpose

This workshop develops intuition for transient heat flow: how thermal diffusivity sets the timescales and length scales over which thermal signals propagate, how half-space and plate cooling models describe the thermal evolution of oceanic lithosphere, and how latent heat modifies cooling in Stefan problems.

Learning Outcomes

Students will:

- Understand how thermal diffusivity controls timescales.
- Compare thermal and chemical diffusion processes.
- Interpret half-space and plate cooling models.
- Use Fourier series to conceptualize thermal evolution.

Session Flow (225 min)

0–10 min: Warm-up: Wedge vs. Orogen

Both accretionary wedges and orogens form at convergent margins, but they experience very different thermal histories. Before we build a detailed model for one of them later today, predict how surface heat flow evolves over the life of each.

Accretionary wedge

Sediment and crust are continuously scraped off a subducting oceanic plate and stacked at the trench. The wedge is built from material that has spent little time near the surface – it arrives cold, wet, and is rapidly buried by whatever is scraped off next.

- While subduction is active, is heat flow through the wedge likely to be higher or lower than a stable craton? What is actually supplying heat to this material, and what is working against it staying warm?
- Once subduction stops – no more cold material arriving, no more rapid burial – what happens to heat flow? Does it recover to the same level as before subduction started, or could it overshoot?

Orogen

Continental collision thickens the crust – doubling it is not unusual. The thickened crust contains more total radiogenic material than before, buried deeper than it used to be.

- Right after thickening, is the crust hotter or colder than it will eventually become? (Think about how long it takes buried heat production to actually raise temperatures at depth – you'll compute this properly in Activity 9.A.)
- Once the orogen has had a long time to adjust, how does its eventual heat flow compare to a stable craton – higher, lower, or about the same?

Sketch both

On the axes below, sketch your predicted heat flow through time for *both* settings, on the same plot. The dashed line marks typical cratonic heat flow, for reference.



We will not resolve the wedge case in detail today – keep your prediction in mind as a point of contrast. The orogen case is exactly what Activity 9.B (Collisional Thermal Evolution) works through properly, including why the answer to the first orogen question is more surprising than it looks.

10–25 min: Mini-lecture — The diffusion equation and thermal diffusivity

- The heat (diffusion) equation: $\partial T / \partial t = \kappa \nabla^2 T$.
- Thermal diffusivity $\kappa = k / (\rho c_p)$: what controls it, typical values for crustal rocks.
- Scaling argument: thermal length $z \sim \sqrt{\kappa t}$; thermal timescale $t \sim z^2 / \kappa$.
- Key difference from steady state: temperature *evolves* in time.

25–70 min: Activity 9.A: Thermal Diffusivity, Thermal Length, and Timescales

- Review $\kappa = k / (\rho c_p)$.
- Emphasize modest variation in crustal rocks.
- Contrast sediments, rock, and ice.
- Penetration depth scaling: $z \sim \sqrt{\kappa t}$.
- Borehole paleothermometry vs ice-core records.
- Depth–resolution tradeoffs.

70–100 min: Activity 9.B: Thermal vs Chemical Diffusion

- Both obey $\sqrt{Dt}$ scaling.

- Chemical diffusion often multi-component.
- Zoning persistence vs thermal smoothing.

100–110 min: Break**110–145 min: Activity 9.C: Half-Space Cooling**

- Assumptions and boundary conditions.
- Growth of thermal boundary layer.
- Oceanic lithosphere as a thermal clock.

145–175 min: Activity 9.D: Plate Cooling and Fourier Series

- Fixed plate thickness as boundary condition.
- Fourier series as superposition of thermal modes.
- Mode decay times and geological interpretation.

175–205 min: Activity 9.E: Stefan Problems in Geology

- Latent heat as effective heat capacity.
- Cooling times of dikes vs sills.
- When analytic solutions break down.

205–215 min: Wrap-up

Bring the session together. Ask: *What single quantity controls how fast heat moves through rock, and what are its typical values?* Revisit the timescale scaling $t \sim z^2/\kappa$ and ask students to give one geological example from each activity where this scaling is evident. Highlight the progression from the simple half-space to the plate model to the Stefan problem as increasing physical realism.

215–225 min: Preview: Finite Differences

Analytic solutions to the diffusion equation require simplifying assumptions: constant properties, simple geometries, and idealized boundary conditions. Real geological systems violate these. Next week we develop finite difference methods, which replace the differential equation with a local energy balance that can be applied to arbitrarily complex geometries and boundary conditions. You will first derive the finite difference equations from first principles and then solve the heat equation yourself by hand—becoming the computer.

Workshop Notes

Workshop Notes

Activity 9.A

Collisional Thermal Evolution

Purpose

The warm-up asked you to predict how heat flow evolves over the life cycle of an accretionary wedge and an orogen. This activity works through the orogen case using a real published thermal model: a thickened, initially hot crust relaxing toward a *cooler* equilibrium, what that means for the thermal history of an individual rock parcel, and what real evidence exists for how long this relaxation actually takes.

Learning Outcomes

By the end of this activity, you should be able to:

- Predict how a perturbed (non-equilibrium) geotherm relaxes toward a lower-heat-flow equilibrium state, and identify which depths change fastest and slowest.
- Construct a qualitative pressure-temperature-time (P-T-t) path for a rock parcel undergoing decompression and cooling.
- Interpret real proxy data for the timescale of post-collisional thermal relaxation, and connect it to the steady-state assumption used elsewhere in this course.

Background: Fischer et al.'s Thermal Model

Fischer et al. model collisional thermal relaxation as a crust cooling from a hot initial state to a cooler final one:

“... cool from a geotherm parameterized by initial surface heat flow (q_0) and crustal heat production (A) to an infinite time geotherm with lower heat flow (q_f) but the same A value... Initial thermal lithospheric thicknesses are 60–90 km and final thicknesses are 190–230 km... Conductivity in all cases is $2.6 \text{ W m}^{-1} \text{ }^\circ\text{C}^{-1}$, and mantle potential temperature is $1,300 \text{ }^\circ\text{C}$ with an adiabatic gradient of $0.3 \text{ }^\circ\text{C km}^{-1}$.”

This activity uses her $q_0 = 70 \text{ mW/m}^2$ and $q_f = 45 \text{ mW/m}^2$ (the middle of her three q_f cases), $A = 0.7 \text{ } \mu\text{W/m}^3$ (held fixed, per her model), and a radiogenic layer 45 km thick – a value not stated in her caption, chosen here because it reproduces her stated lithospheric thickness ranges almost exactly (71 km initial, 210 km final) when both states are treated as steady-state geotherms using the same method as your Week 8 activities.

A caveat worth being explicit about: holding A fixed is a simplification. Part of why q_f ends up lower than q_0 in real orogens is erosion – stripping away radiogenic upper crust reduces the heat production actually available, not just the temperature. Fischer's model doesn't separate that from pure thermal relaxation; it just prescribes a lower q_f as the net outcome. Keep both mechanisms in mind, even though the model itself only shows you one.

Tasks

Question 1. Predicting the Relaxation

- (a) Two mechanisms could both make q_f lower than q_0 : the crust losing its initial excess heat by conduction, and erosion removing radiogenic material from the upper crust. Fischer's model only represents one of these explicitly (holding A fixed) – which one, and what would the other actually change in the equations if you tried to include it?

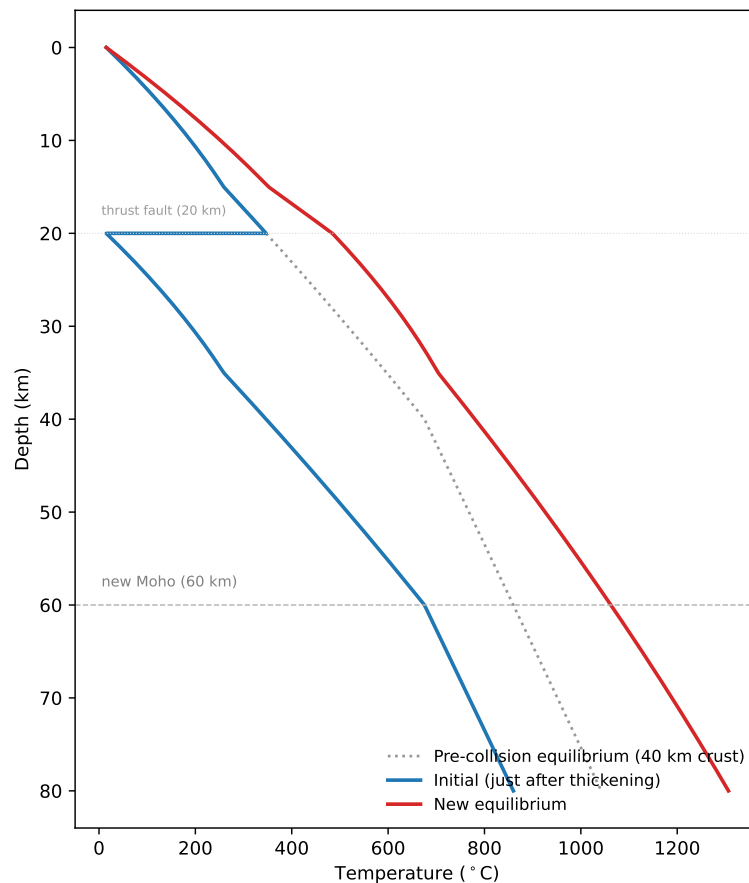


Figure 1: Initial (red, $q_0 = 70 \text{ mW/m}^2$) and final (blue, $q_f = 45 \text{ mW/m}^2$) steady-state geotherms, both with $A = 0.7 \text{ } \mu\text{W/m}^3$ over a 45 km radiogenic layer, following Fischer et al.'s parameterization.

- (b) On Figure 1, sketch how you'd expect the temperature at 30 km and at 100 km to evolve between the initial (red) and final (blue) curves. Which of your two depths do you expect to change faster, and why?

- (c) What would you need to know to say *how long* this relaxation actually takes, in years? (You don't need to compute it – just state what's missing.)

Question 2. A P-T-t Path

Consider a rock parcel at 35 km depth in the initial (hot, q_0) state, which is then exhumed by erosion at a roughly constant rate, reaching 15 km depth by the time the column has relaxed most of the way to the final (q_f) state.

- (a) Using your answer to Question 1, does this parcel's temperature rise, fall, or do both over this interval? (Careful: it is losing overburden *and* sitting in a column that is itself cooling – do both effects push temperature the same direction?)

- (b) Sketch this parcel's path on a plot of pressure (proportional to depth) vs. temperature.

- (c) This captures only the decompression-and-cooling half of the full path a real orogenic rock experiences – metamorphic petrology typically teaches a full *clockwise loop*, with an earlier burial-and-heating (prograde) stage this simplified two-endpoint model doesn't include. Why might a model that starts already-thickened and already-hot be a reasonable simplification for studying the relaxation timescale specifically, even though it skips how the rock got there?

Question 3. Real Evidence: Root Decay Through Time

Fischer et al. compiled R – as best can be determined without the original paper's full methods, the ratio of surface relief (topographic height relative to the surrounding plain, not elevation above sea level) to crustal root thickness – for orogens of many different ages. For a fixed root, R tracks the buoyancy that root is actually providing per unit thickness – so as a root cools and densifies toward normal mantle density, the same root thickness supports less relief, and R falls, even before the root itself has shrunk.

- (a) Does R decay smoothly with age, or does it flatten out at some point the way plate-cooled oceanic lithosphere does?

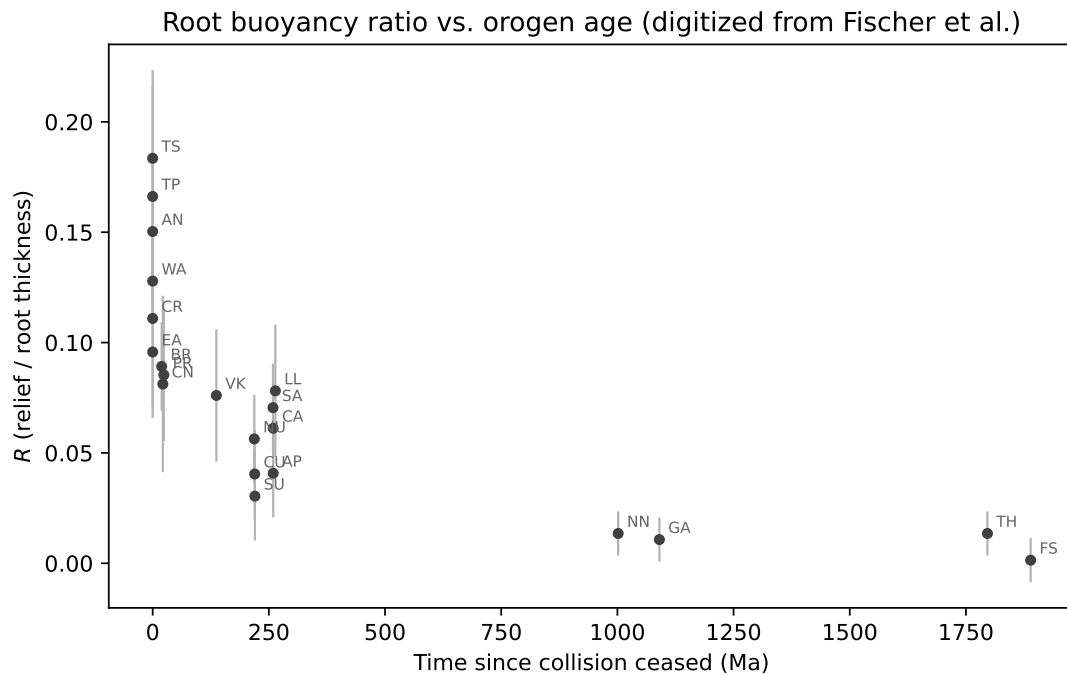


Figure 2: R vs. time since collision ceased, digitized from Fischer et al., all ages shown. Error bars are per-point estimated uncertainties (one-sided estimates, applied symmetrically).

- (b) Week 8's problem set computes thermal isostasy *assuming* steady state. Based on these figures, roughly how old does an orogen need to be before that assumption is reasonable?

- (c) Both this activity's model and the oceanic cooling activity describe a column relaxing away from an initial thermal anomaly. Given that both relax via the same kind of physics, would you expect $R(t)$ to follow the same functional form as oceanic $q(t)$, oceanic $s(t)$, neither, or is there not enough information in this figure alone to tell?

Question 4. Estimating Heat Flow from R – [Placeholder]

This section is still in development. Fischer's own $q_0 = 70 \text{ mW/m}^2$ and $q_f = 45 \text{ mW/m}^2$ are natural calibration anchors for the youngest and oldest ends of the R curve – the calibration points earlier

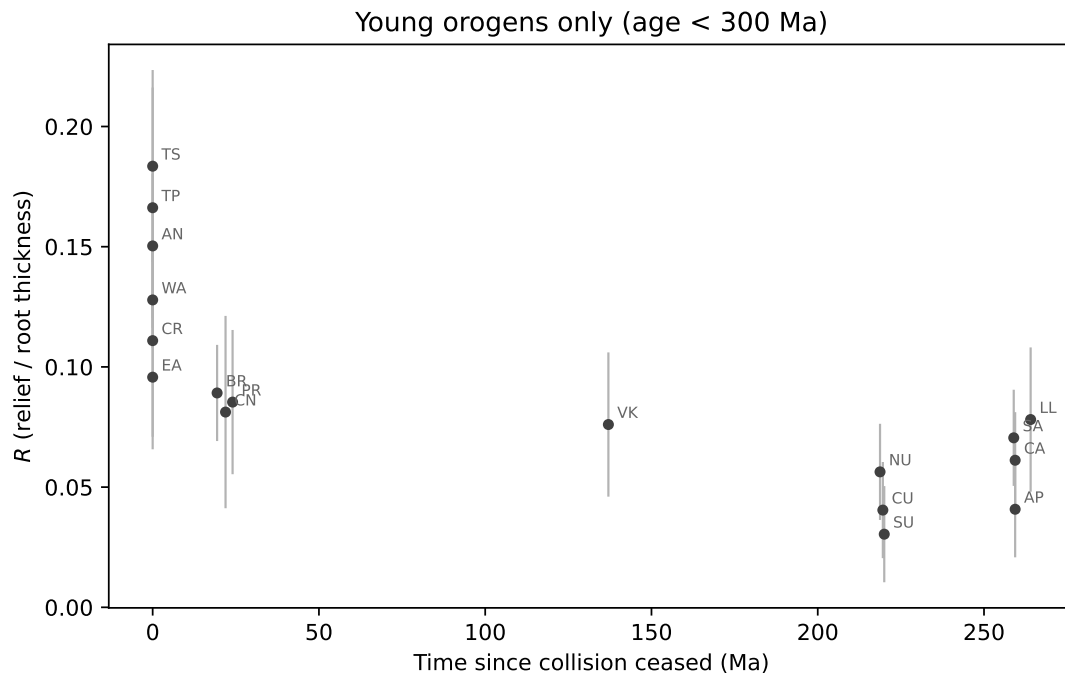


Figure 3: Same data, young orogens only (age < 300 Ma).

identified as missing may now be available. Plan: use these anchors to convert $R(t)$ into an estimated heat-flow-through-time curve for collisional orogens, the collisional counterpart to the oceanic figure from the last activity.

Key Ideas

A thickened column's surface heat flow can fall as it relaxes, not just rise. Which direction depends on the mechanism: burying radiogenic crust deeper (trapping its heat) raises the eventual equilibrium; losing an initial excess of heat – whether from syn-collisional heating, magmatism, or simply not yet having conducted away – lowers it. Real orogens likely involve both, plus erosion actively removing radiogenic mass, which Fischer's fixed- A model doesn't separately represent.

The same relaxation shows up as two different observables, on two different timescales of confidence. Temperature evolution is what the physics predicts directly; R (relief per unit root thickness) is what can actually be measured, for real orogens, over geological time. Neither alone tells the whole story.

A P-T-t path is a rock's biography, not a snapshot. Everything else in this course has asked "what does the geotherm look like right now" – a P-T-t path asks what a single parcel experienced on its way to wherever it ended up. This activity only shows half of it (decompression and cooling); the full clockwise loop taught in metamorphic petrology also includes how the rock got buried and hot in the first place.

Activity 9.B

Thermal Diffusivity, Thermal Length, and Timescales

Purpose

To understand transient heat flow in terms of:

- how far thermal signals propagate in a given time,
- how long it takes thermal changes to affect a given depth,
- and how the *thermal length scale* controls signal shape and timing.

Learning Outcomes

By the end of this activity, you should be able to:

- Define thermal diffusivity and explain which material properties control it.
- Use the thermal length scale $\ell_T = 2\sqrt{\kappa t}$ to estimate how far a thermal signal has propagated or how long it takes to reach a given depth.
- Interpret the error function solution and identify where temperature changes are largest.
- Compare thermal diffusion timescales for different materials and connect these to real geological processes, including how long a thermal event at a given crustal or lithospheric depth takes to be felt elsewhere.

Background

Thermal diffusivity is defined as

$$\kappa = \frac{k}{\rho c_p},$$

where k is thermal conductivity, ρ is density, and c_p is specific heat.

Material	Thermal Conductivity k (W m ⁻¹ K ⁻¹)	Thermal Diffusivity κ (m ² s ⁻¹)	Thermal Diffusivity κ (km ² My ⁻¹)
Ice (glacial / ice sheet average)	1.6–2.0	$(0.7\text{--}0.9) \times 10^{-6}$	22–28
Clay-rich sediments	0.6–1.5	$(0.2\text{--}0.5) \times 10^{-6}$	6–16
Granite	2.5–3.5	$(0.8\text{--}1.2) \times 10^{-6}$	25–38
Basalt / Gabbro	1.7–2.2	$(0.6\text{--}1.0) \times 10^{-6}$	19–32
Peridotite (upper mantle)	3.3–4.5	$(1.0\text{--}1.5) \times 10^{-6}$	32–47

Values are representative ranges. Thermal conductivity and diffusivity vary with temperature, porosity, mineralogy, fluid content, and fabric. Diffusivities are calculated using $\kappa = k/(\rho c_p)$ and rounded for scaling arguments rather than precise modeling.

For diffusion-dominated heat transfer, the characteristic depth over which temperature changes propagate scales as

$$z \sim 2\sqrt{\kappa t}.$$

This thermal length sets both:

- the depth to which temperature changes have propagated after time t ,
- the timescale required for a thermal perturbation at depth to influence shallower levels.

Error Function Solution

For instantaneous heating or cooling at a boundary, the temperature evolution is given by the error function solution:

$$T(z, t) = T_s + (T_b - T_s) \operatorname{erf}\left(\frac{z}{2\sqrt{\kappa t}}\right).$$

Important: the argument of the error function,

$$\eta = \frac{z}{2\sqrt{\kappa t}},$$

controls the *shape and timing* of the thermal signal.

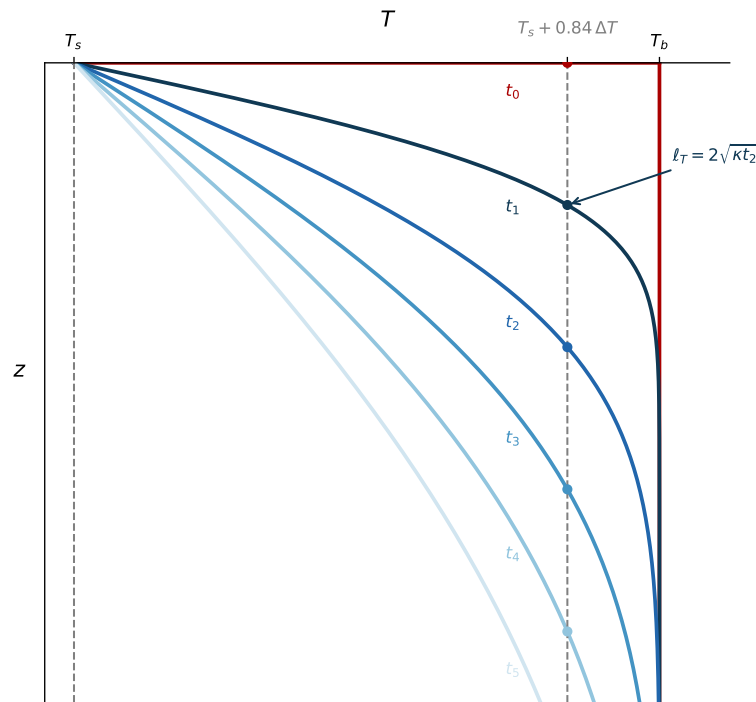


Figure 4: Half-space cooling at three times, $t_1 < t_2 < t_3$. Each curve reaches $T_s + 0.84 \Delta T$ (dashed line) at its own thermal length $\ell_T = 2\sqrt{\kappa t}$ – the same fractional temperature change, at a depth that grows with time.

Tasks

Question 1. Rock vs. Ice

Using the diffusivity table, compute the thermal length $\ell_T = 2\sqrt{\kappa t}$ for rock (use granite, $\kappa \approx 30 \text{ km}^2/\text{My}$) and for ice ($\kappa \approx 25 \text{ km}^2/\text{My}$) at each timescale below. The first row is worked for you.

t (yr)	t (My)	ℓ_T , rock (km)	ℓ_T , ice (km)
10^3	10^{-3}	0.35	0.32
10^5			
10^7			

(a) Describe how the depth of thermal influence differs between rock and ice at each timescale.

- (b) Explain why borehole paleothermometry in rock requires much greater depths to recover long-term climate signals than ice-core records – is it mainly a difference in κ , or something else?

Question 2. Distance Given Time

- (a) For a fixed time t , describe how temperature varies with depth z as the value of η increases.

- (b) Explain why temperature changes are largest for $\eta \ll 1$ and become negligible for $\eta \gg 1$.

- (c) Using Figure 4, how does the depth at which 84% of ΔT has been reached change between t_1 , t_2 , and t_3 ? Does it scale the way you'd expect from $\ell_T = 2\sqrt{\kappa t}$?

Question 3. Time Given Distance: How Long Does a Thermal Event Take to Be Felt?

Rearranging the thermal length relation,

$$t \sim \frac{z^2}{4\kappa},$$

gives the timescale for heat to travel a distance z . Using $\kappa \approx 30 \text{ km}^2/\text{My}$ (representative crust/mantle value), complete the table below. The first row is worked for you.

Depth	z (km)	t (My)
Mid-lower crust	20	3.3
Moho	35	
LAB	100	

- (a) Consider a plume impinging on the base of the lithosphere. Using your table, roughly how long would it take before this thermal perturbation could influence surface temperatures?

- (b) Following delamination of a dense crustal root, heat is supplied from below. Using your table, explain why crustal heating is delayed rather than instantaneous – and delayed by roughly how long?

Question 4. Interpreting “Arrival” of Heat

- (a) Does heat arrive at a specific time, or does it diffuse gradually? Use the error function shape in Figure 4 to justify your answer.

- (b) For $\eta \approx 1$, what fraction of the total temperature change has occurred? What does this imply about defining a single thermal “response time”?

- (c) Discuss why the concept of a sharp thermal front is inappropriate for diffusion-dominated heat transport.

Geological Interpretation

- Thick lithosphere acts as a strong thermal filter, delaying surface response to basal heating.
- Thermal signals generated at depth are attenuated and spread out in time as they propagate upward.

- The thermal length scale controls which geological processes can be observed at the surface, and on what timescales.

Key Idea

In diffusion problems, time and distance are interchangeable through the thermal length $\ell_T = 2\sqrt{\kappa t}$. Heat does not “arrive” at a depth at a single moment; instead, the error function describes a gradual, scale-dependent thermal response controlled by diffusivity, distance, and boundary conditions. A thermal perturbation at the Moho takes millions of years to be felt at the surface; one at the base of the lithosphere takes tens of millions – diffusion is slow enough that most crustal thermal events are, for practical purposes, invisible at the surface on human timescales.

Activity 9.C

Oceanic Lithosphere: Half-Space and Plate Cooling

Purpose

Newly formed oceanic lithosphere cools as it moves away from a ridge, and that cooling leaves two independent, measurable fingerprints: surface heat flow and seafloor depth. This activity derives both from the half-space cooling solution, shows why each becomes linear in a particular transformed variable, and tests the model against real global compilations – including where it breaks.

Learning Outcomes

By the end of this activity, you should be able to:

- Derive surface heat flow and subsidence from the half-space cooling solution $T(z, t)$.
- Explain why q^{-2} is linear in age and why depth is linear in $\sqrt{\text{age}}$, for a half-space.
- Identify where real oceanic data does and does not follow this prediction, and connect any departure to the plate cooling model's finite basal boundary condition.
- Relate the timescale at which half-space and plate predictions diverge to the characteristic time $\tau = L^2/\kappa$.

Background

The half-space cooling solution, with $T(0, t) = T_0$ and $T(z, 0) = T_m$ for all $z > 0$, is

$$T(z, t) = T_0 + (T_m - T_0) \operatorname{erf}\left(\frac{z}{2\sqrt{\kappa t}}\right).$$

This is the same solution from Activity 9.A, now applied with T_0 as the seafloor (bottom water) temperature and T_m as the mantle temperature it cools from.

Aside: oceanic lithosphere technically forms by seafloor spreading, the $\gamma = 1$ limit of the rifting model in your notes and problem set – that limit is singular in the pure-shear formulation and requires a separate treatment, given in your notes. We do not need it here; the half-space and plate solutions below are self-contained.

Tasks

Question 1. Deriving Surface Heat Flow

Using $q_0(t) = k \left. \frac{\partial T}{\partial z} \right|_{z=0}$ and the chain rule, along with $\frac{d}{dx} \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} e^{-x^2}$, derive an expression for $q_0(t)$.

You should arrive at $q_0(t) = \frac{k(T_m - T_0)}{\sqrt{\pi\kappa t}}$.

Question 2. Deriving Subsidence

Thermal contraction of the cooling column, balanced isostatically against the water that fills the space it vacates, produces subsidence

$$s(t) = \frac{\rho_m}{\rho_m - \rho_w} \alpha \int_0^\infty [T(z, 0) - T(z, t)] dz.$$

- (a) Write $T(z, 0) - T(z, t)$ in terms of the complementary error function, $\text{erfc}(\eta) = 1 - \text{erf}(\eta)$.

- (b) Substitute $\eta = z/(2\sqrt{\kappa t})$ to change variables in the integral. You will need the identity $\int_0^\infty \text{erfc}(\eta) d\eta = \frac{1}{\sqrt{\pi}}$ (given – this one isn't worth re-deriving here).

- (c) Combine your results to arrive at $s(t)$.

You should arrive at $s(t) = \frac{2\rho_m\alpha(T_m - T_0)}{\rho_m - \rho_w} \sqrt{\frac{\kappa t}{\pi}}$.

Question 3. Why These Linearize

- (a) Using your result from Question 1, show that q_0^{-2} is linear in t .

- (b) Using your result from Question 2, in what variable is $s(t)$ linear?

Question 4. Estimating the Fit Constants by Hand

Write the half-space predictions as $q(t) = c_q/\sqrt{t}$ and $s(t) = c_s\sqrt{t}$, absorbing all the physical constants ($k, \kappa, T_m - T_0, \rho_m, \alpha, \dots$) into the single constants c_q and c_s . Two real data points from each dataset are given below; s is depth relative to the ridge-depth reference of 2500 m.

Heat flow		Bathymetry		
t (Ma)	q (mW m ⁻²)	t (Ma)	depth (m)	$s = \text{depth} - 2500$ (m)
19.8	114.0	11.3	3633	
51.3	72.1	41.3	4633	

- (a) Using $q(t) = c_q/\sqrt{t}$, compute c_q from each heat flow point separately. Are your two estimates consistent with each other?

- (b) Using $s(t) = c_s\sqrt{t}$, compute c_s from each bathymetry point separately. Are your two estimates consistent with each other?

- (c) Which pair of estimates agreed more closely – c_q or c_s ? Given what generates scatter in young oceanic heat flow, suggest why one dataset might let a small number of points pin down its constant more reliably than the other.

Question 5. Real Data: Heat Flow

- (a) How does the full-dataset fit, $c_q \approx 549$, compare to your two hand estimates from Question 4? Does this change how much you'd trust a constant estimated from just one or two points?

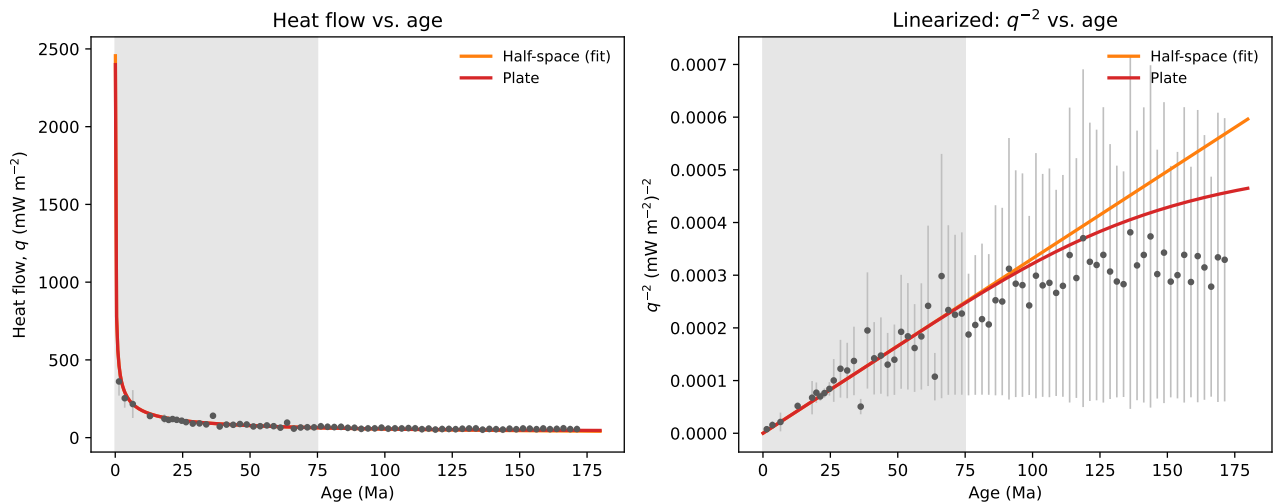


Figure 5: Sediment-corrected global heat flow vs. seafloor age (binned, error bars show one standard deviation). The half-space curve here is fit by least squares to *every* point in the shaded region ($t < 75$ Ma), giving $c_q \approx 549 \text{ mW m}^{-2} \text{ Ma}^{1/2}$.

- (b) At young ages, does the data sit above, below, or on the half-space prediction? What physical process, not included in this conductive model, might explain that?

- (c) At the oldest ages shown, do half-space and plate predictions still agree closely, or have they begun to separate? What does that tell you about how old the seafloor needs to be before the two models are distinguishable in heat flow alone?

Question 6. Real Data: Bathymetry

- (a) How does this full-dataset fit compare to your two hand estimates from Question 4? Is the improvement from using more points as large as it was for heat flow?

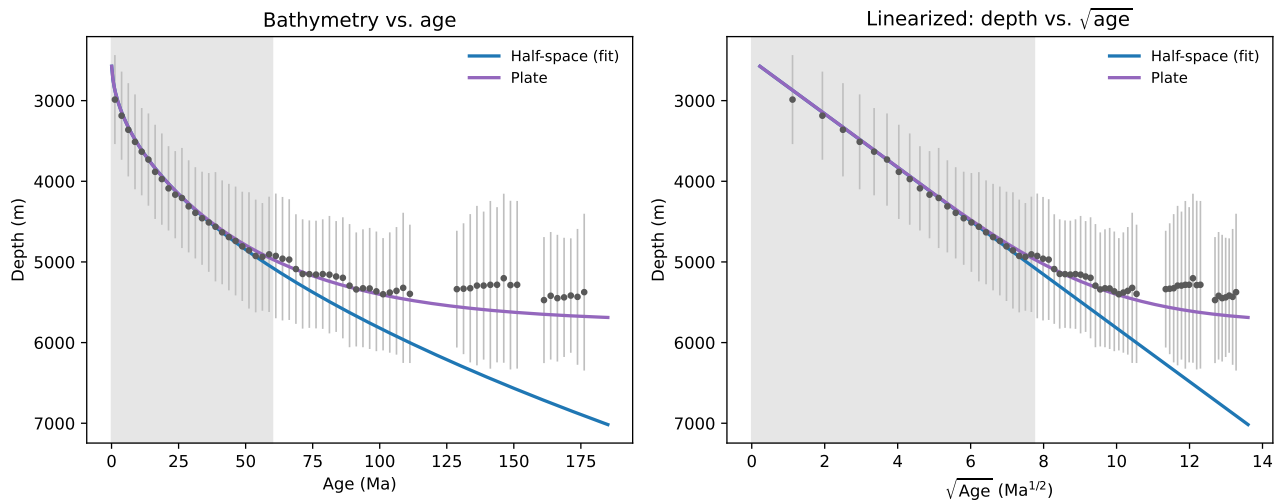


Figure 6: Global bathymetry vs. seafloor age (binned, error bars show one standard deviation). The half-space curve is fit to every point in the shaded region ($t < 60$ Ma), giving $c_s \approx 332 \text{ m Ma}^{-1/2}$.

- (b) Compare this figure to Question 5's heat flow figure. In which observable does the half-space model's failure become obvious sooner, as a function of age?

- (c) The half-space model has no plate thickness L in it at all – it's a model of infinite depth. Explain, physically, why real subsidence eventually stops following the half-space prediction.

Question 7. Early and Late Time

The characteristic time for the plate model is $\tau = L^2/\kappa$.

- (a) For $t \ll \tau$, the plate solution reduces to the half-space solution. Why does a finite-thickness plate behave like an infinite half-space when it's young?

- (b) For $t \gg \tau$, heat flow approaches a constant, $q_0 \rightarrow k(T_m - T_0)/L$ – independent of t . Why does the plate's finite thickness impose a floor on how thin the thermal boundary layer can get, when a half-space's doesn't?

Key Ideas

The same solution predicts two independent, testable observables. Heat flow comes from the temperature gradient at the surface; subsidence comes from integrating the temperature deficit through the whole column. Both derive from the same $T(z, t)$, but they linearize in different variables (t vs. $\sqrt{t}$) and, as you saw, they reveal the half-space model's breakdown at different ages.

A model with no built-in length scale cannot predict its own failure. The half-space solution has no plate thickness in it, so nothing internal to the model signals when it stops applying – that only becomes visible by comparing to real data, or to a model (plate cooling) that does have a length scale.

Early-time agreement between two models doesn't guarantee late-time agreement. The plate and half-space solutions coincide for $t \ll \tau$ by construction, not by coincidence – and diverge once the finite basal boundary condition is actually felt, at $t \sim \tau$.